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The changing language and sentiment of conversations about climate change in Reddit posts over sixteen years

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Global environmental change has been a topic of discussion in the media for many decades, and social perception of media terminology has been a topic of research interest. However, a systematic review of large-scale online discussions and the terminology used has not been undertaken. Here, we analyze 16 years of Reddit discussions, encompassing 11.5 billion posts, to examine how language surrounding climate change has evolved over time from 2005 to 2021. We applied sentiment analysis, polarity, subjectivity, and readability metrics to discussions of "global warming" and "climate change". We found that the use of "climate change" surpassed "global warming" in 2013, with "climate change" associated with more negative sentiment and higher subjectivity. Additionally, we observed a decline in the proportion of climate-related discussions over time despite the increasing total number of posts. These findings suggest that public engagement with climate topics on Reddit is waning, and the choice of terminology significantly influences the tone and complexity of the discourse. Our results have important implications for how climate issues are communicated and perceived by the public.

Global environmental change has been discussed in the media for many decades, with varied issues attention[1] and different framing from country to country[2]. Climate journalism has emerged and evolved as a field[3].

Global warming and climate change are terms that are sometimes used interchangeably[4,5]. Yet, they do have different semantic contexts and connotations: Global warming refers to a long-term increase in global temperatures, while climate change denotes changes in climate patterns, in particular involving the increased levels of carbon dioxide in the atmosphere[6].

Climate change has been used in academic papers since the fifties[7], but it has been more widely popularized in the public discourse by Frank Luntz, a Republican communication strategist, who in 2002 advised the Bush administration to use it instead of global warming, as a less frightening term[8], and to make the perceived lack of scientific certainty a vital issue in the debate about the environment. Although we do not address how this intentional shift in nomenclature influenced the discourse prior to 2005 since the dataset began in late 2005, this would be an interesting area for further research.

Even though for liberals, global warming and climate change are associated equally with rising temperatures and, overall, the heat impact on the planet, conservatives see climate change as more disconnected from it[9]. It is also worth noting that the US media, in particular, are prone to relativization of the topic and frequently evoke scientific uncertainty and controversy regarding human-induced climate change[5]. Whether the Republican plan was successful or not, over the years, a noticeable shift in vocabulary from global warming to climate change in the popular press and academic papers alike occurred[10,11].

The social perception of these terms on the Internet has been studied, e.g., by Google Trends[12,13]. There have been also many studies of the perception of climate change and environmental challenges on the social media, with a heavy bias towards Twitter/X[6,14,15], possibly because of the ease of data access.

However, it has not been the subject of a systematic review of large-scale online discussions. It is essential, though, as online discussion fora provide a platform for contrasting arguments and reasoning, in contrast to Twitter/X, which relies on short, linguistically restricted statements, and in contrast to just Google Trends, which only reflects people's search interests. Online discussions are also a good proxy for general public interest and perceptions and may help to understand the emergence of public spheres in climate-related topics[16].

Reddit is the largest and longest-running platform for discussions in English, and using it allows for analyzing critical social changes and trends[17,18]. As of 2023, Reddit, with its unique structure centered around interest-based communities known as subreddits, hosts over 850 million monthly active users (MAUs)[19] and 2.1 billion unique visitors[20], engaging in in-depth discussions and sharing a wide array of content. Notably, it is the 5th most visited social media platform globally[21], with approximately 48% of

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visits (Tiago Bianchi, 2024) originating from the United States. Reddit represents approximately one-third as much social media traffic as Twitter/X which serves as a hub for real-time updates and public discourse (Dixon, 2024). Although Facebook remains the dominant force with just over 3 billion monthly active users, catering to a broad demographic by facilitating personal connections and diverse content consumption of which approximately 24% are based in India and 12% are based in the United States (Dean, 2024), Reddit's model of specialized, user-moderated subreddits is particularly resonant among specific interest groups, steadily growing in its impact on public opinion and media trends.

In our study, we perform the first large-scale analysis of 11.5 billion posts in regard to climate change and global warming.

Results and discussion

Our analysis confirms the lexical transition from global warming to climate change indeed took place, albeit a decade later than the Republican initiative purportedly instigating this shift. This change was abrupt, with climate change becoming the dominant term in the discussions analyzed around 2013, as illustrated in Fig. 1. This shift in terminology is further corroborated by Google search trends that reveal a parallel change occurring around two years later, as shown in Fig. 2. These findings suggest a broad adoption of the term climate change in both digital discourse and search behaviors during this period, reflecting broader societal and political shifts.

Furthermore, extending our investigation into Google Trends data from 2004 to 2021, well beyond those previously analyzed in studies focusing on searches before 2014[6,22], we found a similar shift in terminology from climate change to global warming around 2015 beyond Reddit. This alignment between Reddit discussions and Google search trends, despite the slight difference in timing, supports the consistency of this lexical evolution across various Internet platforms. Additionally, the six-month rolling averages for the sum of the two Google trend lines, which approximately represent the overall climate-related searches over time, show a strong correlation (Spearman correlation coefficient of 0.89) with the percentage of Reddit discussions about climate-related topics (see Fig. 2). This correlation further suggests that the trends observed in Reddit discussions are reflective of broader trends across Internet-based media, underscoring the pervasive shift in how climate issues are discussed online.

Although some prior studies viewed the terms somewhat interchangeably in traditional mass media from 1995 to 2004[23], more recent studies suggest important differences. For example, one study found that the term global warming elicited lower levels of belief in the phenomenon, especially among Republicans, while the term climate change elicited higher levels of belief, particularly among the same group[24]. Another study found that red states (i.e., more conservative states in the United States) tended to prefer the term global warming while blue states (less conservative) tended to prefer the term climate change[25]. More recently, one study found that on Twitter/X global warming was more closely associated with human causation and conveyed more of an immediate and direct impact, whereas climate change appeared to be a broader term that could be framed as a natural phenomenon, potentially downplaying the necessity for human intervention[26]. Another study found that tweets mentioning climate change were more likely to be associated with environmental and political content, whereas tweets using global warming were more likely to be linked to weather (including heatwaves) and energy, and the authors suggested that

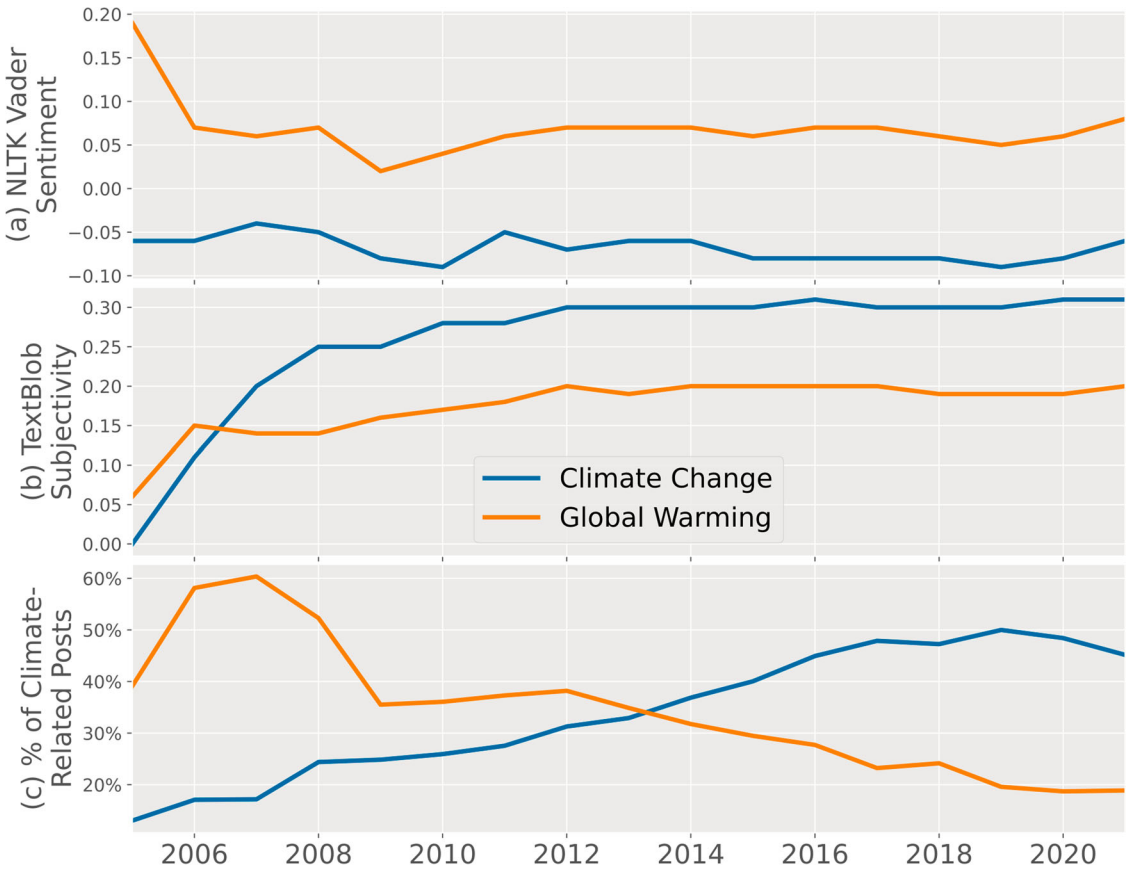

Fig. 1 | Changes in the climate-related discourse on Reddit. This figure shows the frequency of usage and associated sentiment and subjectivity of the terms "global warming" and "climate change" in Reddit posts over time. **a** The bottom section illustrates the decline in the use of "global warming" and the corresponding rise in "climate change" since 2005. **c** The top section depicts the sentiment associated with "climate change" becoming more negative over time, while **b** the middle section highlights that the subjectivity of both terms has increased, with "climate change" consistently being discussed in more subjective language compared to "global warming" since 2005. *File*: Fig. 1 Climate Related NGram Properties by Year 2005 to 2021.svg.

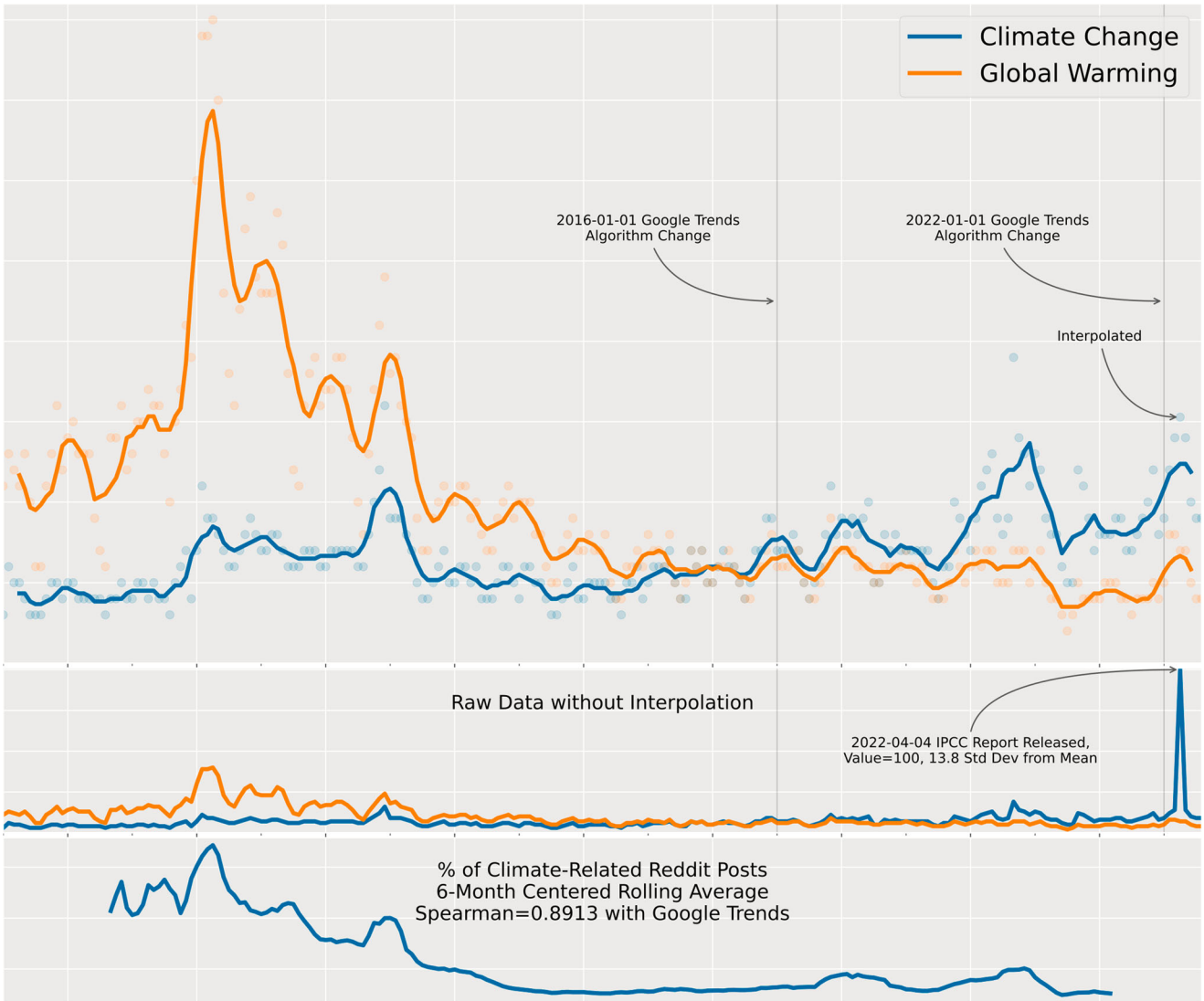

Fig. 2 | Google Search Trends. This figure shows the Google search term trends for climate change and global warming for the 18 years from January 2004 to August 2022. The top panel shows the 6-month rolling average lines with the light dots delineating the non-averaged data points with the exception of the month of the 2022 IPCC report, where an interpolated point (mean of 5 surrounding points) was used to mitigate the exceptional outlier shown in the middle panel. The bottom panel shows the 6-month rolling average of the relative percentages of all climate-related posts, namely the blue line in Fig. 5. Vertical lines represent the dates on which Google implemented algorithmic changes to how it tracks searches. In April 2022, the month during which the IPCC released its most recent and most alarming report, there was a large spike in searches reported; however, we are unable to ascertain if the two are related. *File*: Fig. 2 Google Search Trend Scores from 2004 to 2022 for Climate Change and Global Warming.svg.

terminology may influence public perception and engagement[15]. A 2014 survey of 1657 adults found that people reported using and hearing global warming far more often than climate change—45% of the time vs. 12% and 35% vs. 16%, respectively[22]; although the term preferences are consistent with our study's observed shift in Reddit and Google Search Trends around that time, the considerable difference between the perceived frequencies and those observed suggests there may be important underlying factors that require further investigation.

The analysis of the TextBlob subjectivity scores revealed a significantly higher level of subjectivity in posts pertaining to climate change (Fig. 1) suggesting that individuals discussing this term tend to express their perspectives in a less objective manner. Conversely, discussions labeled under global warming were characterized by a tone suggesting objective factuality. This distinction may imply that contributors discussing climate change are potentially more vigilant in differentiating between opinion and fact. Such an observation is aligned with expectations and underscores a critical discourse dynamic: it appears that those referring to global warming may be more likely to perceive their viewpoints as unequivocal truths. This distinction in discourse styles not only illuminates differing perceptions but also potentially influences the public and policy discourse surrounding climate issues. In short, the choice of terms may significantly impact how the underlying phenomenon has been received.

Our analysis challenges prevailing assumptions by demonstrating that the term global warming is associated with more positive sentiment compared to climate change. An average positive sentiment was consistently observed over all time periods analyzed for global warming, whereas climate change corresponded with predominantly negative scores (Fig. 1). This disparity may suggest that individuals discussing global warming are less apprehensive about climatic changes than those using the term climate change, possibly reflecting a delay in adopting the newer terminology. Notably, the subreddit askscience, known for its scientific rigor, displayed the most positive sentiments for both terms and had some of the highest composite scores (Fig. 3). Contrasting with a 2015 study[6], which utilized Semantria® to analyze Twitter/X sentiments over two months,

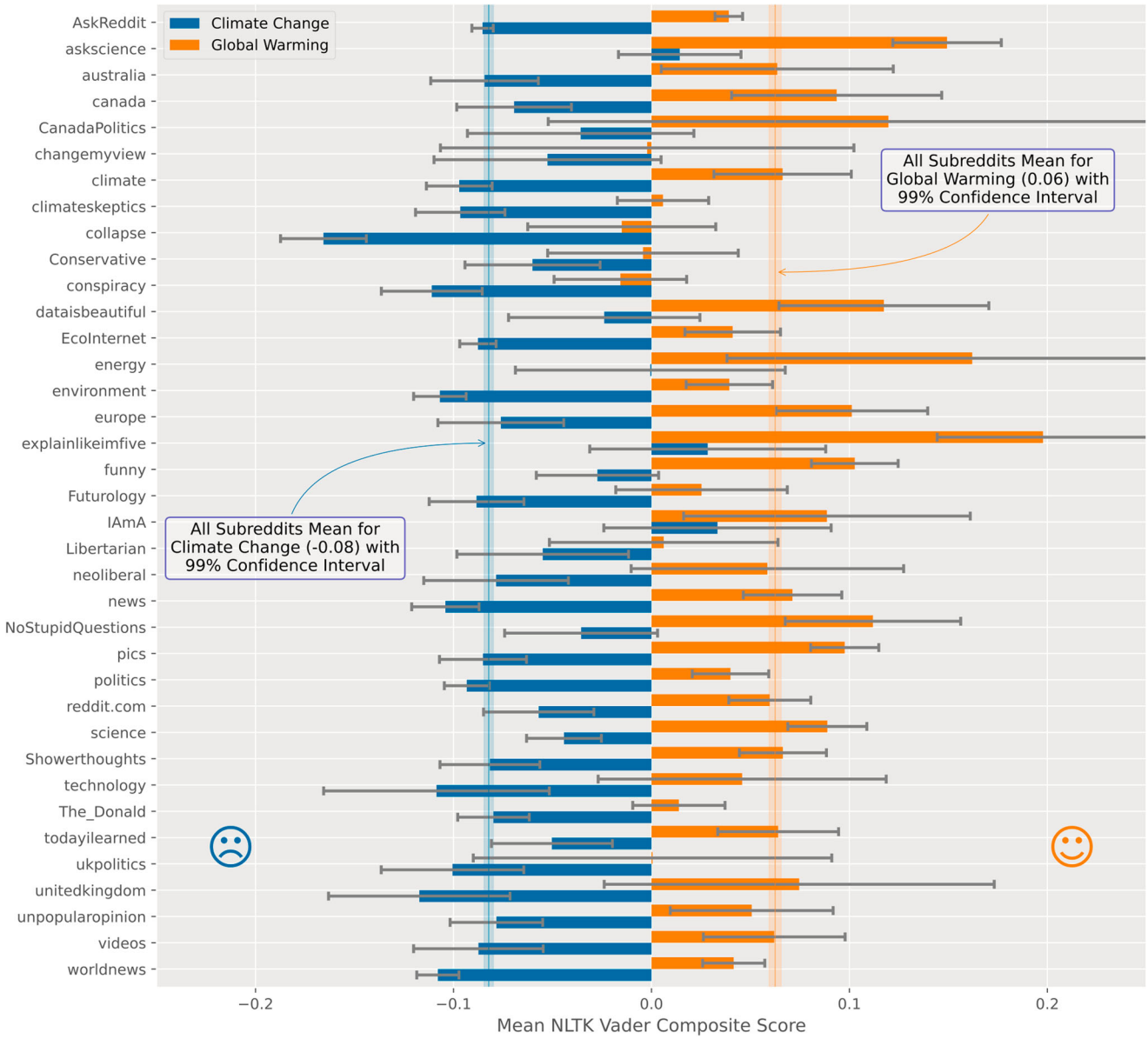

Fig. 3 | Mean NLTK Vader composite scores by subreddit. This figure depicts the mean NLTK Vader Sentiment Composite Score for climate-related posts containing the terms "climate change" or "global warming" across all subreddits, which are plotted as vertical lines with the shaded area representing the 99% confidence interval while the bars represent the mean for each of these for a given subreddit with the error bars showing the 99% confidence interval for that given subreddit and term. The subreddits (37 total) were selected by choosing the 50 highest-volume subreddits with more than 100 climate-related posts. *File*: Fig. 3 Mean Vader Composite by Subreddit.svg.

our broader assessment over 15 years using VADER sentiment scores uncovered an opposing trend, aligning more closely with sentiment self-report survey results from 2014[22]. This incongruence underscores the need for further exploration into the dynamics shaping sentiment. It is notable that during Donald Trump's first presidency, there was a modest decrease in both sentiment polarity and subjectivity within discussions of climate change and global warming. Contributions during this period trended toward more negative tones and were phrased more as declarations of fact than expressions of personal viewpoints. The underlying reasons for this shift in discourse style remain unclear and warrant further investigation.

As illustrated in Fig. 3, the term climate change consistently registers less favorable Vader sentiment scores compared to global warming across various subreddits, regardless of the spectrum of views on climate validity presented by these forums (such as The Conservative, The_Donald, and conspiracy versus climate, askscience, and science). This pattern indicates that the more positive sentiment typically associated with global warming transcends political and scientific divides. Notably, subreddits dedicated to educational content, including askscience, explainlikeimfive, and IAmA, tend to adopt more positive tones in their discussions. This consistency suggests a broad, enduring perception gap between the two terms.

We observed that posts containing the term climate change exhibit a small but statistically significant increase in the use of complex vocabulary across all subreddits. However, this trend does not consistently manifest within individual subreddits or across different forums (refer to Fig. 4). The subreddit aimed at changing the original poster's (OP) views displayed the highest usage of complex words in discussions related to climate topics. Paradoxically, the subreddit explainlikeimfive, which is intended to simplify complex subjects for easier understanding, had the second-highest average use of complex vocabulary in its posts.

When considering the proportion of Reddit posts addressing climate-related topics, a noticeable decline is observed. From 2005 to 2010, up to 1.4%

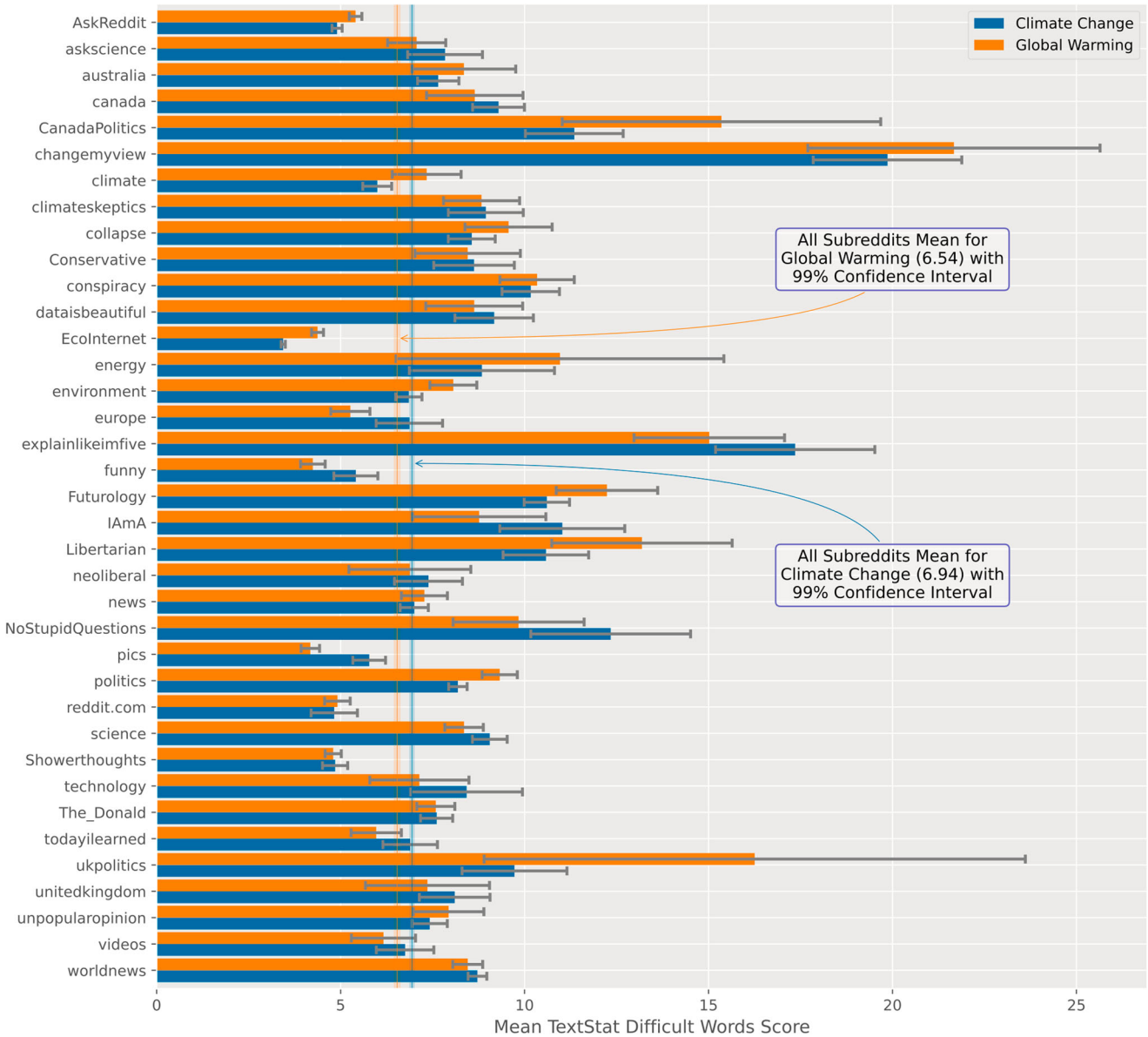

Fig. 4 | Mean TextStat difficult words scores by subreddit. In this figure, the mean TextStat Difficult Words score for all posts containing the terms "climate change" and "global warming" across all subreddits are plotted as vertical lines with the shaded area representing the 99% confidence interval while the bars represent the mean for each of these for a given subreddit with the error bars showing the 99% confidence interval for that given subreddit and term. The subreddits (37 total) were selected by choosing the 50 highest-volume subreddits with more than 100 climate-related posts. *File*: Fig. 4 Mean TextStat Difficult Words Score by Subreddit.svg.

of discussions were climate-related. However, this percentage has since shown a steady decline, ultimately averaging around 0.1% of posts by 2021 (Fig. 5). This apparent decrease may be somewhat misleading due to the overall growth in Reddit's popularity, which could disproportionately affect the perceived share of climate discussions. In reality, while the absolute number of climate-related discussions has continued to increase, the expansion of conversations on other topics has outpaced them. Despite the actual increase in climate discussions, this trend suggests a relative decrease in engagement with climate issues within the Reddit community, serving as a subtle indicator that the topic may be diminishing in proportional popularity.

As an additional observation, our analysis of word difficulty revealed that the ecointernet subreddit utilized the least complex language in discussions pertaining to climate change, global warming, and climate topics in general (see Fig. 4). Conversely, the Changemyview subreddit, which often hosts debates and confrontational dialogs, exhibited the most complex sentence structures, likely due to a reliance on academic sources and technical language. Surprisingly, the Explain to Me Like I'm 5 (explainlikeimfive) subreddit, which aims to simplify concepts, employed more complex vocabulary than other forums

Our findings deepen the existing literature by offering a comprehensive analysis of Reddit discussions, contrasted against broader search term trends, thereby enhancing our understanding of the evolution of climate discourse across diverse online platforms.

Conclusions

Global environmental change is a highly politicized and ideologically charged topic within public discourse[27,28]. The mainstream media have been systematically covering this issue for years, often disproportionately highlighting contrarian views over expert opinions[29]. Despite extreme weather events drawing considerable media attention[30], overall media coverage on climate change has grown[31,32]. Moreover, even the more conservative outlets have started acknowledging the burning problem[33]. It is important to note that climate-related topics may not have the same level of politicization in all regions of the world[34].

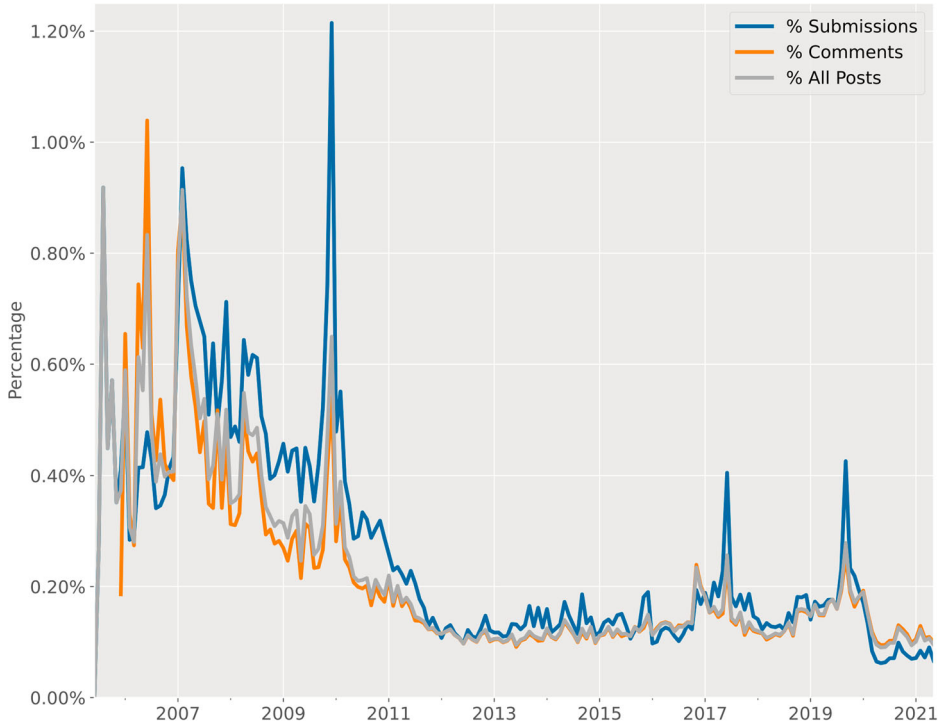

Fig. 5 | Percentage of all posts related to climate change. This figure shows the relative frequency of climate-change-related topics in time as a percentage of discussions on Reddit. The discussion frequency has been trending down over the time period analyzed. *File*: Fig. 5 Percentage of All Reddit Posts Matching Climate-Change Related Terms.svg.

Unlike the growing media attention, our analysis of large-scale online discussions reveals a different trend. Over 16 years of Reddit posts show a decreasing proportional interest in climate-related topics. While the absolute number of climate discussions has increased, the expansion of other topics on Reddit has outpaced it, suggesting a relative decline in engagement with climate issues. This trend might be attributed to various factors, including topic fatigue, competition from other crises[35], and the emergence of other engaging topics.

Our study highlights significant linguistic and sentiment differences between the terms global warming and climate change. Discussions about global warming tend to use less sophisticated language and exhibit more positive sentiments compared to those about climate change. This finding suggests that the term global warming is perceived less negatively and is communicated in a more accessible manner. This linguistic distinction is crucial as it can influence public perception and engagement with the topic.

Furthermore, our analysis indicates that online discussions precede Google searches in reflecting the language shift from global warming to climate change. This suggests that social media platforms like Reddit can serve as early indicators of changing public attention and discourse. Monitoring these platforms may provide more immediate insights into shifts in public interest and sentiment than traditional search trend analyses.

Our findings have notable implications for policymakers, communicators, and climate advocates. The decreasing proportional interest in climate discussions on Reddit highlights the challenge of maintaining public engagement on this critical issue. The language used to discuss climate change can affect how the topic is perceived and engaged with by the public. Therefore, strategic communication efforts should consider these linguistic nuances to effectively mobilize public interest and action.

Moreover, our study underscores the importance of addressing the potential gap between media coverage and public engagement. While the media continues to amplify climate issues, translating this attention into sustained public concern and action remains a challenge. Policymakers and communicators must develop innovative strategies to reinvigorate public interest and participation in climate discussions, ensuring that the urgency of global environmental change is not lost amid other competing issues.

In conclusion, the window for mobilizing public attention around global environmental change may be closing. The observed trends suggest that informal discussions on social media platforms substantially influence the broader debate among non-experts. To address the climate crisis effectively, it may be imperative to leverage these platforms for proactive and engaging communication strategies. Understanding and harnessing the dynamics of online discourse can play a crucial role in shaping public perception and driving collective action toward mitigating global environmental change.

Further investigation into how discussions about climate change are shifting across various online platforms could provide a more comprehensive understanding of the evolution of public discourse on this critical issue.

Methods

Data collection

We obtained a large dataset ($n = 11.486 \times 10^9$) containing Reddit.com posts archived from the PushShift.io data repository[36] for the years 2005 through June 2021 for both Reddit Submissions (RS files) and Reddit Comments (RC files). Submissions are the top-most post that initiates a discussion within a group known as a subreddit while comments are replies to either a submission or another comment. We uncompressed each monthly file, and removed duplicates. We then used a series of regular expressions to identify all posts related to climate change while ignoring all instances of political, economic, financial, academic, work, social, job, or employment climate, as well as terms such as "climate control" used to refer to topics not related to climate change while still retaining all posts that contained explicit references to climate change, global warming, and related climate specific topics such as references to the Paris Accord, the Kyoto Protocol, Greta Thunberg, and others.

This resulted in 15.341×10^6 posts likely to be related to climate change (0.134%). These posts were then converted from JSON format to a more data science-friendly CSV format retaining all the relevant information on which we performed language detection using both the polyglot[37] and langdetect Python libraries, keeping only posts that were determined to be in English by both ($n = 1.501 \times 10^6$, 97.9% of matched posts). We subsequently concatenated and lemmatized the title and body of all the remaining posts

using standard NLTK library methods (nltk.stem.WordNetLemmatizer) and removed all English stop words[38]. Each post was then analyzed using the VaderSentiment[39].

VADER (Valence Aware Dictionary and sEntiment Reasoner): A lexicon and rule-based sentiment analysis tool that is specifically attuned to sentiments expressed in social media. VADER uses a combination of a sentiment lexicon and a set of heuristic rules to estimate sentiment scores for textual data[39].

VADER's rule-based approach, which takes into account both the semantic meaning of words and the context in which they are used (including punctuation, capitalization, and modifiers), makes it highly effective for the nuanced understanding of sentiment in social media posts. This approach provides a robust framework for analyzing how sentiment regarding climate change and global warming is expressed in online discussions, contributing significantly to the study's findings on public engagement and perception related to these topics.

Sentiment Score: VADER analyzes the text to calculate a compound score that indicates the overall sentiment. This compound score is a normalized, weighted composite score that ranges from -1 (extremely negative) to 1 (extremely positive). This scoring is beneficial in understanding the general sentiment of each post regarding climate change or global warming.Top of Form

TextBlob: Provides a simple API for common natural language processing (NLP) tasks such as part-of-speech tagging, noun phrase extraction, sentiment analysis, classification, translation, and more. In this study, we utilized TextBlob's sentiment analysis features to assess the polarity and subjectivity of the posts.

- Polarity: This refers to the emotional orientation of the words used in the text. Polarity is a float that lies between −1.0 and 1.0, where −1.0 indicates highly negative sentiment, 0 indicates neutrality, and 1.0 indicates a highly positive sentiment. It essentially measures the degree to which the language used in a text leans towards a positive or negative emotion. This metric is crucial for understanding the general sentiment expressed in discussions about climate change and global warming on social media platforms.
- Subjectivity: This measures the amount of personal opinion and factual information contained in the text. The value ranges from 0 to 1, where 0 is very objective, and 1 is highly subjective. Subjective text expresses personal feelings, views, or beliefs, whereas objective text provides factual information. Analyzing subjectivity helps in distinguishing between posts that are based on personal opinions and those based on objective reports, which is vital for evaluating how discussions on climate change are framed by the public.

In our analysis, these metrics enabled a deeper understanding of the emotional and subjective layers within the climate change discourse on Reddit, providing insights into how people emotionally engage with the topic and the extent to which discussions are opinion-driven or fact-based[40].

TextStat's Difficult Words analysis is a tool used to quantify the complexity of vocabulary in a text. It evaluates the readability of content by counting words not found on a predefined list of commonly understood words, typically based on the Dale-Chall word list[41]. This metric is particularly useful in assessing the accessibility of text to audiences with varying levels of literacy. In our study, we employed TextStat Difficult Words analysis to measure the linguistic complexity of discussions within Reddit posts concerning 'climate change' and 'global warming.' By applying this analysis, we aimed to determine the sophistication of language that could potentially influence the comprehensibility and reach of the discourse on these environmental issues. The insights gained help to understand how the complexity of language in online discussions may affect public engagement and awareness.

We then analyzed the unigram, bigrams, and trigrams and subsequently analyzed the occurrences of the "climate" unigram when not followed by "change" and the "climate change" bigram as well as the "global" unigram when not followed by "warming" and the "global warming" bigram. We accomplished this by replacing all permutations of occurrences of "climate change" with "climate change" and "global warming" with "global warming" after lemmatization and performing a unigram analysis. For each of the resultant unigrams, we analyzed sentiment, polarity, and readability subdivided by the top 40 subreddit, and looked at the relative trends for the unigrams across each year.

Selecting climate-related posts

To ensure our analysis focused solely on content relevant to climate change, we developed a set of criteria to filter posts from the PushShift.io data repository. This involved using regular expressions to select posts that specifically mentioned climate change-related terms and exclude irrelevant contexts.

Methodology summary: Our approach was twofold. First, we included posts that contained explicit references to climate-related topics, such as "global warming," "carbon emissions," and specific environmental policies (e.g., the Paris Accord). For example, one of our expressions

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\\bglobal\\w*[\W_]*(? : warm\\w*|cool\\w*|temper\\w*|heat\\w*|hot[estr]*\\b|cold[estr]*\\b)
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captures discussions around global temperature changes.

Second, we excluded posts where the term "climate" appeared in non-environmental contexts, such as "political climate" or "climate control," using negative look behind assertions in our regular expressions. For instance, the expression (?<!politic\\w*)\\bclimat ensures the word "climate" is not preceded by "political" or related modifiers.

Inclusion and ethics statement

This study utilizes a global dataset of publicly available Reddit posts to capture a wide-ranging perspective on climate change discourse. All analyses were conducted at an aggregate level, focusing on patterns in public discussions rather than individual user data. The authors acknowledge the importance of global representation in environmental discourse and have made efforts to minimize biases in data selection, aiming to reflect a broad spectrum of perspectives on climate-related topics.

Data availability

All data that support the findings of this study have been deposited in FigShare and are available via FigShare at https://doi.org/10.6084/m9.figshare.26828467. The dataset includes all the raw and processed data used in the analyses as well as the data generated to create all figures and plots presented in the study. The code used for analysis and figure generation has also been deposited in the same repository and can be found in the Jupyter Notebook and Colab Notebook compatible file titled "Published Reddit General Climate Related Plots 2005-2021.ipynb" in the Code folder.

Code availability

All code used to analyze the data and create all figures is freely available in live format on Google Colab (Reddit Climate Analysis 2005–2021), via FigShare, and on GitHub complete with a detailed readme to allow recreation of all pre-processed data.

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Author contributions

Gabriele Fariello was responsible for data acquisition, including sourcing and organizing datasets from PushShift.io and AcademicTorrents.com, and performed all data preprocessing steps such as data cleaning, regular expression scripting, language detection, and text normalization. Gabriele conducted all subsequent analyses, including sentiment analysis, subjectivity analysis, and readability assessments, using tools such as VADER, TextBlob, and TextStat. He also generated all the figures and plots in the manuscript and led the manuscript's writing. Dariusz Jemielniak supervised the project, contributing to its conceptualization, including the

integration of Google Trends data, and supporting literature searches. Dariusz reviewed the analysis methodology, provided critical feedback, and contributed substantively to interpreting results and refining the manuscript's framing.

Competing interests

The authors declare no competing interests.

Additional information

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